



Installation and Commissioning Instructions

Automation Marine
mtu NautIQ CoOperate

E532910/00E



A Rolls-Royce
solution

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Preface

Welcome to the Rolls Royce Power Systems AG *Installation and Commissioning Instructions mtu NautIQ CoOperate*.

Intended Audience for This Document

The *Installation and Commissioning Instructions mtu NautIQ CoOperate* is intended for users with mechanical, electronic, and networking knowledge. The mtu NautIQ CoOperate must be installed by a qualified marine electronics technician. For electrical installations, the installer must comply with the NMEA 0400 Installation Standard. The installer must also comply with ISO/TR 10134:2020,

Small craft — Electrical devices — Established practices for the design, construction and installation of lightning-protection systems, for lightning protection.

The Commissioning procedure that takes place after installation will be performed by Rolls Royce Power Systems AG personnel. They will assist you in performing the sea trials at the end of the commissioning procedure.

Note: Only qualified personnel who are fully informed of the system safety hazards should install or operate this equipment.

Associated Documents

- *Operating Instructions mtu NautIQ CoOperate E532909/00E*

Book Conventions

This guide contains the following information highlights and cross-references:



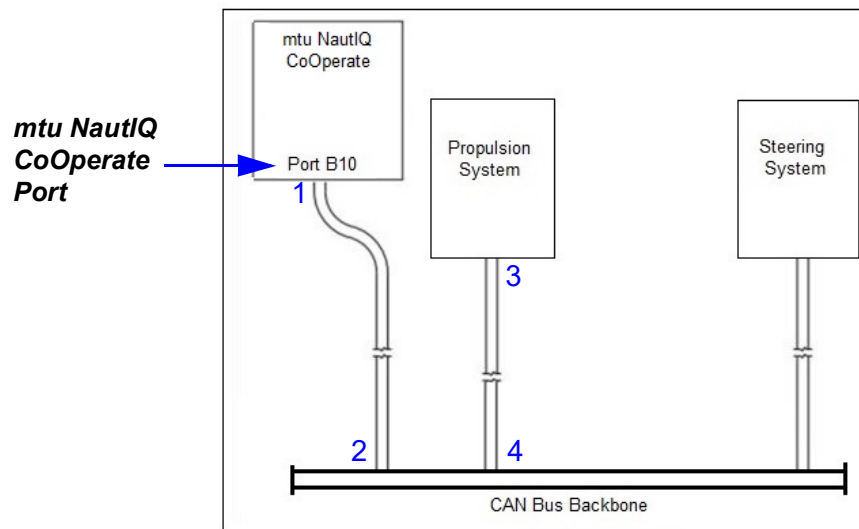
Warning: Warns the user regarding actions that may result in physical damage to the system or personal injury.



Caution: Cautions the user regarding actions that may result in operational issues or data loss.

Note: Identifies important points, helpful hints, special circumstances, or alternative methods.

- Cross references indicate the location of additional information regarding the chosen topic. References may include either headings on specific pages or entire chapters, for example, See “**Introduction**” on page 1-2.
- Buttons, field names, and selections are shown in **bold**.
- Blue boxes, text, and arrows highlight areas of interest on screenshots or images as shown:



Chapter 1: Overview

Topics described in this chapter include:

Introduction	page 1-2
System Overview	page 1-2
mtu NautIQ CoOperate Connections	page 1-6
Installation Overview	page 1-4
Customer-Provided Components	page 1-5
Minimum Installation Requirements	page 1-6
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Introduction

This guide describes how to plan for and install your mtu NautIQ CoOperate system.

System Overview

The mtu NautIQ CoOperate system includes three major components (see **Figure 1-1**):

- mtu NautIQ CoOperate Console
- mtu NautIQ CoOperate User Interface Laptop (mtu NautIQ CoOperate UI)
- mtu NautIQ CoOperate Remote Helm Controller (mtu NautIQ CoOperate Remote Helm)

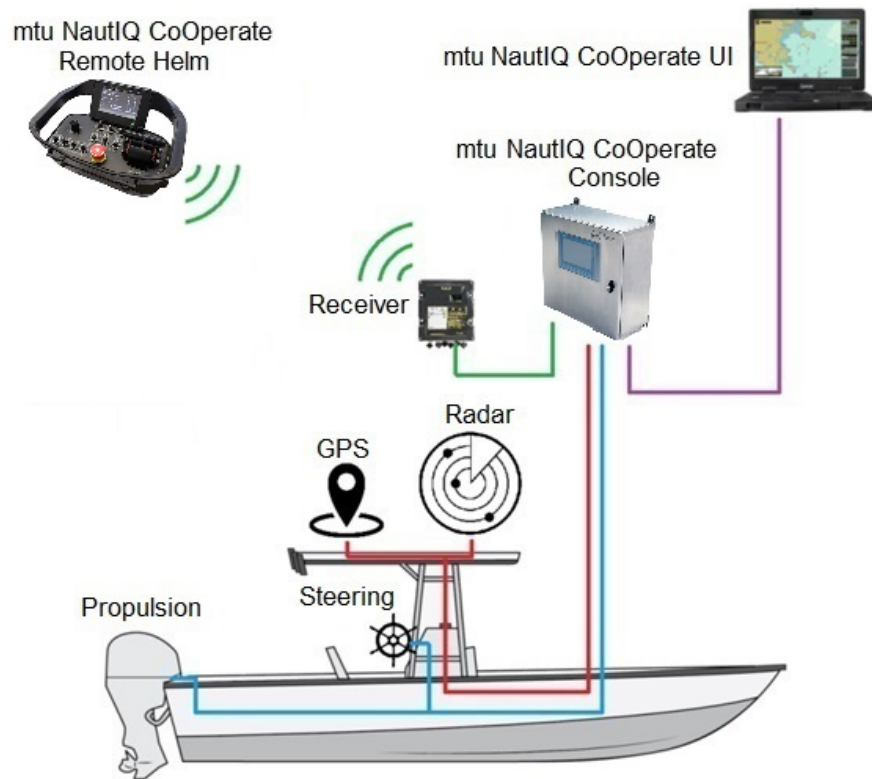
Figure 1-1: mtu NautIQ CoOperate System



mtu NautIQ CoOperate Connections

Figure 1-2 shows an overview of the mtu NautIQ CoOperate connections.

Figure 1-2: mtu NautIQ CoOperate Connections

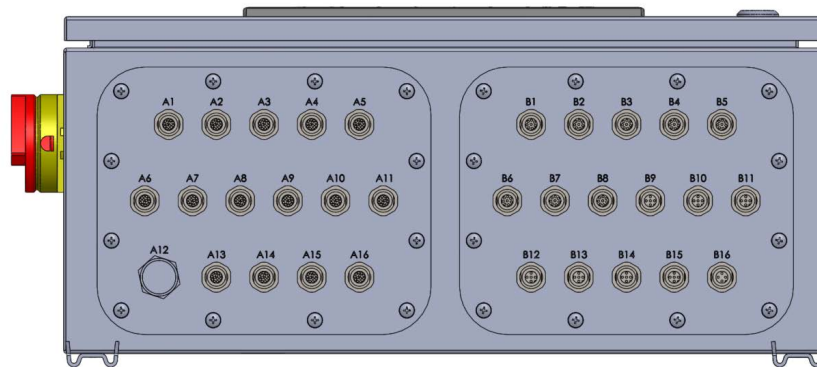


The mtu NautIQ CoOperate Console is mounted within your vessel. (The mtu NautIQ CoOperate Console is designed to be installed in a dry and environmentally protected space within the vessel. If there is no option to install within such a space, the mtu NautIQ CoOperate Console should be installed in a location that protects it from adverse weather conditions, water spray, and direct sunlight, and it should be mounted high enough to avoid any standing water.) The mtu NautIQ CoOperate UI connects to the Console via an Ethernet connection (either hardwired or over a wireless solution provided during installation). The mtu NautIQ CoOperate Remote Helm communicates to the mtu NautIQ CoOperate via a receiver that connects to the Console.

The vessel's steering and propulsion systems and all the vessel's onboard electronics, including the GPS, radar, and other auxiliary equipment all connect to the mtu NautIQ CoOperate Console.

Figure 1-3 shows the connector plate on the bottom of the mtu NautIQ CoOperate Console through which all connections are made.

Figure 1-3: Connection Panel on mtu NautIQ CoOperate Console



Refer to the *mtu NautIQ CoOperate User's Guide* for a full system overview.

Installation Overview

There are three main parts to the mtu NautIQ CoOperate installation:

- Mechanical Installation
- Electrical Installation
- Network Installation

Additionally, after installation, a Commissioning process is performed by a Rolls Royce Power Systems AG engineer or an authorized representative.